Bonding & Properties Lab

How bond type predicts electrical conductivity and melting point

Name: _____ Period: _____ Date: _____

Objective

You will test the electrical conductivity of four substances as solids, liquids, and dissolved in water, and measure each one's melting point. Using your data, you will figure out how the way a substance is bonded (ionic or covalent) predicts both of these properties.

Background

Conductivity: Electric current only flows when charged particles are free to move around. A substance can only conduct electricity if two things are both true: (1) charged particles (ions) are present, and (2) those particles are free to move. Whether both conditions are met can change depending on whether the substance is a solid, a liquid, or dissolved in water.

Melting point: The substances you will test are held together in two different ways — as an ionic lattice, or as a covalent (molecular) solid. The strength of the attractions holding the solid together determines how much energy (and therefore how high a temperature) it takes to melt it.

Data Table

Fill this in as you complete each test in the simulation (or copy it from your completed Data Table tab).

Substance	Solid	Liquid	Aqueous	Melting Point
Sodium Chloride (NaCl)				
Table Sugar / Sucrose ($C_{12}H_{22}O_{11}$)				
Paraffin Wax ($C_{25}H_{52}$)				
Calcium Chloride ($CaCl_2$)				

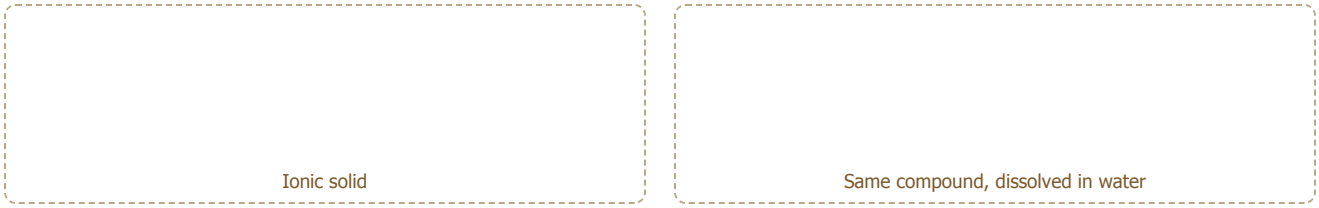
Keep this page for reference — the analysis questions on the next page ask you to use this data.

Analysis Questions

Answer each question in 1–3 complete sentences, using your data table above.

1. What two things must be true for any substance to conduct electricity? Be specific.

2. Ionic compounds conduct electricity when melted or when dissolved in water, but not as a solid. Covalent (molecular) compounds never conduct, in any state. Explain why these two patterns are different. In the boxes below, draw a particle diagram comparing an ionic solid to the same ionic compound dissolved in water — you may use arrows to show which particles, if any, are moving.



3. Pure water (H_2O) is a covalent (molecular) compound. Would you expect pure water to conduct electricity? Explain why or why not.

4. Which would you predict to have the higher melting point: potassium bromide (KBr) or naphthalene ($C_{10}H_8$, the substance mothballs are made of)? Explain using bond type.

5. Which would you predict to have the higher melting point: magnesium oxide (MgO) or iodine (I_2)? Explain your reasoning.

6. Sucrose ($C_{12}H_{22}O_{11}$) dissolves completely in water, yet the solution does not conduct electricity. What does this tell you about what happens to a sucrose molecule when it dissolves?

7. An unknown solid conducts electricity when melted or dissolved in water, but not as a solid. Would you predict this compound to have a high or low melting point? Explain.